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Laser irradiation apparatus, laser irradiation method, and method for manufacturing semiconductor device

Abstract

A laser irradiation apparatus including: a laser light source configured to emit a linearly polarized pulsed laser light; a first half-wave plate rotatably provided in an optical path of the pulsed laser light; a first polarization beam splitter configured to branch the pulsed laser light from the first half-wave plate into a first pulsed light and a second pulsed light; a second polarization beam splitter configured to combined the first pulsed light with the second pulsed light, the second pulsed light, the second pulsed light being delayed from the first pulsed light by using an optical path length difference between the first pulsed light and the second pulsed light; and a first wave plate rotatably provided in an optical path of a combined pulsed light generated by combining the first pulsed light with the second pulsed light at the second polarization beam splitter.

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Background/Summary

INCORPORATION BY REFERENCE

(1) This application is based upon and claims the benefit of priority from Japanese patent application No. 2020-173773, filed on Oct. 15, 2020, the disclosure of which is incorporated herein in its entirety by reference.

BACKGROUND

(2) The present disclosure relates to a laser irradiation apparatus, a laser irradiation method, and a method for manufacturing a semiconductor device.

(3) Japanese Patent No. 6706155 discloses a laser annealing apparatus for forming a polycrystalline silicon thin film. In Japanese Patent No. 6706155, a solid-state laser apparatus emits linear polarization laser light. The linear polarized laser beam is incident on a polarization beam splitter through a half-wave plate. The polarization beam splitter splits the laser light into two light beams. The two light beams split by the polarization beam splitter are combined with each other at the second polarization beam splitter.

SUMMARY

(4) In such a laser irradiation apparatus, it is desired to irradiate the laser light having an appropriate polarization state with an object.

(5) Other problems and novel features will be clarified from the description of this specification and the attached drawings.

(6) According to an embodiment, a laser irradiation apparatus including: a laser light source configured to emit a linearly polarized pulsed laser light; a first half-wave plate rotatably provided in an optical path of the pulsed laser light; a first polarization beam splitter configured to branch the pulsed laser light from the first half-wave plate into a first pulsed light and a second pulsed light; a second polarization beam splitter configured to combined the first pulsed light with the second pulsed light, the second pulsed light being delayed from the first pulsed light by using an optical path length difference between the first pulsed light and the second pulsed light; and a first wave plate rotatably provided in an optical path of a combined pulsed light generated by combining the first pulsed light with the second pulsed light at the second polarization beam splitter.

(7) According to an embodiment, a laser irradiation method including: (a) emitting a linearly polarized pulsed laser light; (b) causing the pulsed laser light to be made incident on a first half-wave plate rotatably placed; (c) branching the pulsed laser light from the first half-wave plate into a first pulsed light and a second pulsed light; (d) combining the first pulsed light with the second pulsed light, the second pulsed light being delayed from the first pulsed light by using an optical path length difference between the first pulsed light and second pulsed light; (e) causing a

combined pulsed laser to be made incident on a first wave plate rotatably provided in an optical path of the combined pulsed laser, the combine pulsed laser being generated by combining the first pulsed light with the second pulsed light at the second polarization beam splitter.

(8) According to an embodiment, A method for manufacturing a semiconductor device, including the steps of: (S1) forming an amorphous film over a substrate; and (S2) annealing the amorphous film so that the amorphous film is crystalized and a crystallized film is thereby formed, wherein the (S2) annealing comprises the steps of: (A) emitting a linearly polarized pulsed laser light; (B) causing the pulsed laser light to be made incident on a first half-wave plate rotatably placed; (C) branching the pulsed laser light from the first half-wave plate into a first pulsed light and a second pulsed light; (D) combining the first pulsed light with the second pulsed light, the second pulsed light being delayed from the first pulsed light by using an optical path length difference between the first pulsed light and second pulsed light; (E) causing a combined pulsed laser to be made incident on a first wave plate rotatably provided in an optical path of the combined pulsed laser, the combine pulsed laser being generated by combining the first pulsed light with the second pulsed light at the second polarization beam splitter.

(9) According to the embodiment, it is possible to irradiate the laser light having an appropriate polarization state with an object.

(10) The above and other objects, features and advantages of the present disclosure will become more fully understood from the detailed description given hereinbelow and the accompanying drawings which are given by way of illustration only, and thus are not to be considered as limiting the present disclosure.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 shows an optical system of a laser annealing apparatus according to the present embodiment;

(2) FIG. 2 shows an optical system of the laser annealing apparatus;

(3) FIG. 3 shows a laser light split by a polarization beam splitter **33**;

(4) FIG. 4 shows time waveform of the laser light combined by a polarization beam splitter **61**;

(5) FIG. 5 illustrates the polarization state of the laser light synthesized by the polarization beam splitter **61**;

(6) FIG. 6 is a simplified cross-sectional view of a configuration of an organic EL (Electro-Luminescent) display;

(7) FIG. 7 is a cross-sectional view showing a process in a method for manufacturing a semiconductor device according to this embodiment; and

(8) FIG. 8 is a cross-sectional view showing a process in a method for manufacturing a semiconductor device according to this embodiment.

DESCRIPTION OF EMBODIMENTS

(9) A laser irradiation apparatus according to this embodiment is, for example, a laser annealing apparatus that forms an LTPS (Low Temperature Poly-Silicon) film. A laser irradiation g apparatus, a laser annealing method, and a method for manufacturing a semiconductor device according to this embodiment will be described hereinafter with reference to the drawings.

(10) (Optical System of a Laser Irradiation Apparatus)

(11) A configuration of a laser irradiation apparatus according to the present embodiment will be described with reference to FIG. 1. FIG. 1 schematically shows an optical system of the laser irradiation apparatus **1**. A silicon film **101** is formed over the upper surface (the main surface) of a substrate **100**. The laser irradiation apparatus **1** irradiates the silicon film **101** formed over the substrate **100** with laser light **L1**. In this way, it is possible to convert an amorphous silicon film (a-

Si film) **101** into a poly-crystalline silicon film (polysilicon film: p-Si film) **101**. The substrate **100** is, for example, a transparent substrate such as a glass substrate. The substrate **100** is an object to be irradiated with the laser beam.

(12) Note that, an XYZ three-dimensional orthogonal coordinate system is shown in FIG. **1** to clarify the description. The Z-direction is the vertical direction and perpendicular to the substrate **100**. The XY-plane is a plane parallel to the surface of the substrate **100** over which the silicon film **101** is formed. The X-direction is the longitudinal direction of the rectangular substrate **100**, and the Y-direction is the lateral direction of the substrate **100**. Further, in the laser irradiation apparatus **1**, the silicon film **101** is irradiated with the laser light **L1** while the substrate **100** is conveyed in the X-axis positive direction by a conveying mechanism (not shown in FIG. **1**). Note that, in FIG. **1**, the silicon film **101** that has not yet been irradiated with the laser light **L1** is referred to as an amorphous silicon film **101a**, and the silicon film **101** that has already been irradiated with the laser light **L1** is referred to as a polysilicon film **101b**.

(13) The laser irradiation apparatus **1** includes a stage **10**, a laser light source **21**, an optical system **20** and a detection unit **22**. The substrate **100** is disposed over the stage **10**. The stage **10** is a flotation stage (a flotation unit) that floats (i.e., levitates) the substrate **100** by ejecting air. But the stage **10** is not limited to the flotation stage. For example, the stage **10** may be a vacuum chuck type stage. The stage **10** ejects gas toward the substrate **100** from below the substrate **100**.

(14) Therefore, the substrate **100** is conveyed in the X-axis positive direction in a state in which a minute air gap is formed between the stage **10** and the substrate **100**.

(15) The optical system **20** is an optical system for irradiating the silicon film **101** with laser light **L1** in order to crystallize the amorphous silicon film **101a**. The optical system **20** includes a polarization control unit **30** for controlling the polarization state of the laser beam **L1**. The polarization control unit **30** controls the polarization state of the laser beam **L1**. The detailed configuration of the optical system **20** will be described later.

(16) The optical system **20** is disposed above the substrate **100** (on the Z-axis positive side of the substrate **100**). The laser light source **21** is a pulsed laser light source and generates pulsed laser light. The laser light source **21** is, for example, a solid-state laser apparatus such as a Nd: YAG laser apparatus. Further, the laser light source **21** emits pulsed laser light **L1**. The laser light source **21** emits linearly polarized laser light **L1**. The laser light source **21** is not limited to the solid-state laser apparatus, but may be a semiconductor laser apparatus.

(17) The optical system **20** includes a homogenizer for uniformizing the laser beam **L1**, a condenser lens for condensing the laser beam **L1**, and the like. The laser light **L1** forms a linear irradiation region over the substrate **100**. The irradiation region has a line shape in which the Y direction is a longitudinal direction and the X direction is a lateral direction.

(18) The optical system **20** guides the laser light **L1** emitted from the laser light source **21** to the substrate **100**. The laser light incident on the substrate **100** from the optical system **20** is referred to as a laser light **L2**. The amorphous silicon film **101a** is crystallized by the irradiation of the laser light **L2**. The silicon film **101** is irradiated with the laser light **L2** while the part in the substrate **100** to which the laser light **L2** is applied is changed. A uniform polysilicon film **101b** is formed over the substrate **100** by conveying the substrate **100** in the X-axis positive direction by the conveying mechanism of the stage **10**. Needless to say, the laser light source **21** and the optical system **20** may be moved instead of conveying the substrate **100**. That is, any configuration or the like may be used as long as the irradiation region is scanned by the laser light **L2** by moving the substrate **100** and the annealing optical system **20** relative to each other.

(19) Further, the detection unit **22** is disposed over the substrate **100**. The detection unit **22** is provided in order to photograph (i.e., take an image of) the crystallized polysilicon film **101b**. The detection unit **22** includes a camera for photographing the polysilicon film **101b**, for example. Alternatively, the detection unit **22** may include a spectrometer for measuring the spectrum of the reflected light reflected by the polysilicon film **101b**. Furthermore, the detection unit **22** may

include an illumination light source for illuminating the substrate **100**. For example, the illumination light source generates illumination light for illuminating a region irradiated with the laser light **L2**. The camera detects the reflected light from the region illuminated by the illumination light. Since the detection unit **22** can photograph the region irradiated with the laser beam **L2**, the crystal state of the polysilicon film **101b** can be evaluated. It becomes possible to evaluate uniformity and unevenness of the crystal state of the polysilicon film **101b**.

(20) A polarization control unit **30** controls a polarization state of the laser beam **L1** and adjusts a pulse waveform. Specifically, the polarization control unit **30** controls the time waveform of the pulse laser beam by delaying a part of the pulse laser beam. FIG. **2** is a diagram showing a configuration of the polarization control unit **30**. The polarization control unit **30** includes a half-wave plate **31**, a mirror **32**, the polarization beam splitter **33**, the polarization beam splitter **61**, and a wave plate **62**.

(21) The half-wave plate **31** is rotatably provided in the optical path of the laser beam **L1**. The polarization direction of the laser beam **L1** can be adjusted by rotating the half-wave plate **31** around the optical axis of the laser beam **L1**. That is, the linear polarization direction rotates according to the rotation angle of the half-wave plate **31**.

(22) The laser beam **L1** is incident on the mirror **32** through the half-wave plate **31**. The mirror **32** reflects the laser light **L1** toward the polarization beam splitter **33**. The laser beam **L1** reflected by the mirror **32** is incident on the polarization beam splitter **33**.

(23) The polarization beam splitter **33** branches the laser beam **L1** into two pulse beams **L31** and **L32**. The polarization beam splitter **33** branches the incident light according to the polarization state. Specifically, the polarization beam splitter **33** transmits through the P polarization component and reflects the S polarization component. Therefore, the pulse light **L31** transmitted through the polarization beam splitter **33** becomes linearly polarized light of P polarization. The pulse light **L32** reflected by the polarization beam splitter **33** becomes linearly polarized light of S polarization.

(24) The pulse light **L31** and the pulse light **L32** are combined by the polarization beam splitter **61** at a later stage. Two optical paths are provided between the polarization beam splitter **33** and the polarization beam splitter **61**. The pulse light **L31** propagates in one optical path from the polarization beam splitter **33** to the polarization beam splitter **61**, and the pulse light **L32** propagates in the other optical path from the polarization beam splitter **33** to the polarization beam splitter **61**. An optical path length difference is given between the two optical paths so as to delay the pulse light **L32** from the pulse light **L31**.

(25) Between the polarization beam splitter **33** and the polarization beam splitter **61**, an optical path through which the pulse light **L31** propagates is referred to as a preceding optical path **40**, and an optical path through which the pulse light **L32** propagates is referred to as a delay optical path **50**. The optical path length of the delay optical path **50** is longer than that of the preceding optical path **40**. The pulse light **L31** is defined as a preceding pulse light and the pulse light **L32** is defined as a delay pulse light.

(26) A wave plate **41** is arranged in the preceding optical path **40**. The pulsed light **L31** from the polarization beam splitter **33** passes through the wave plate **41**, and then enters the polarization beam splitter **61**.

(27) A mirror **51**, a mirror **52** and a wave plate **53** are arranged in the delay optical path **50**. The pulse light **L32** from the polarization beam splitter **33** is reflected by the mirror **51** and the mirror **52**. The pulse light **L32** reflected by the mirror **52** is incident on the wave plate **53**. The pulse light **L32** passes through the wave plate **53**, and then enters the polarization beam splitter **61**. Since the mirror **51** and the mirror **52** are arranged in the delay optical path **50**, the optical path length becomes longer than that of the preceding optical path **40**. Thus, the pulse light **L32** entering the polarization beam splitter **61** is delayed more than the pulse light **L31**.

(28) In FIG. **2**, two mirrors **51** and **52** are provided in the delay optical path **50**, but the arrangement and the number of mirrors **51** and **52** may be adjusted according to the optical path length

difference. Of course, an optical element other than a mirror may be used to form the optical path length difference. The delay time may be adjusted by adjusting the positions of the mirrors **51** and **52**.

(29) The polarization beam splitter **61** combines the pulse light **L31** with the pulse light **L32**. The polarization beam splitter **61** transmits the pulse light **L31** and reflects the pulse light **L32**. Thus, the pulse light **L31** and the pulse light **L32** propagate in the same optical axis. In a state where the pulse light **L32** is delayed from the pulse light **L31** by the optical path length difference, the pulse light **L31** and the pulse light **L32** are combined with each other.

(30) For example, each of the wave plate **41** and the wave plate **53** is a half-wave plate. The wave plate **41** may be a third half-wave plate and the wave plate **53** may be a fourth half-wave plate. In the preceding optical path **40**, the wave plate **41** is rotatably arranged. The rotational axis of the wave plate **41** is parallel to the optical axis of the pulse light **L31**. In the delay optical path **50**, the wave plate **53** is rotatably arranged. The rotational axis of the wave plate **53** is parallel to the optical axis of the pulse light **L32**.

(31) The wave plates **41** and **53** are provided to reduce the loss of light at the polarization beam splitter **61**. For example, the polarization beam splitter **61** transmits the P polarization component and reflects the S polarization component. The wave plate **41** is arranged to maximize the transmitted light amount of the pulse light **L31** at the polarization beam splitter **61**. That is, the wave plate **41** is arranged at a rotation angle such that the pulse light **L31** incident on the polarization beam splitter **61** becomes P polarization.

(32) The wave plate **53** is arranged to maximize the amount of reflected light of the pulse light **L32** at the polarization beam splitter **61**. That is, the wave plate **53** is arranged at a rotation angle such that the pulse light **L32** incident on the polarization beam splitter **61** becomes S polarization. For example, reflection by the mirror **51** and the mirror **52** may cause the polarization direction of the pulse light **L32** to rotate. Even in such a case, the loss of the pulse light **L32** in the polarization beam splitter **61** can be prevented by rotating the wave plate **53**. The wave plate **41** and the wave plate **53** may be omitted.

(33) As described above, the polarization beam splitter **61** combines the pulse light **L31** with the pulse light **L32**. Therefore, the preceding pulse light and the delayed pulse light propagate coaxially. Here, the pulse light **L31** and the pulse light **L32** combined by the polarization beam splitter **61** are used as a combined pulse light **L60**. The combined pulse light **L60** includes the preceding pulse light and the delayed pulse light.

(34) The light amount of the pulse light **L31** and the pulse light **L32** will now be described. As shown in FIG. 3, the pulse light **L31** and the pulse light **L32** are linearly polarized light orthogonal to each other. Here, the half-wave plate **31** is disposed in an optical path from the laser light source **21** to the polarization beam splitter **33**. Accordingly, by rotating the half-wave plate **31**, the ratio of the pulse light **L31** to the pulse light **L32** can be adjusted. By rotating the half-wave plate **31**, the polarization direction of the laser beam **L1** is changed.

(35) Therefore, the intensity ratio of the preceding pulse light to the delayed pulse light can be changed according to the rotation angle of the half-wave plate **31**. FIG. 4 is a diagram schematically showing the time waveform of the combined pulse light **L60** including the preceding pulse light **P1** and the delay pulse light **P2**. In FIG. 4, three examples are shown when the angle of the half-wave plate **31** is changed.

(36) The upper part of FIG. 4 shows a time waveform when the intensity of the preceding pulse light **P1** is set to be higher than that of the delay pulse light **P2**. In the middle part of FIG. 4, a time waveform is shown when the intensity of the preceding pulse light **P1** is equal to that of the delayed pulse light **P2**. The lower part of FIG. 4 shows a time waveform when the intensity of the preceding pulse light **P1** is set lower than that of the delay pulse light **P2**. By changing the rotation angle of the half-wave plate **31**, the intensity ratio of the preceding pulse light **P1** and the delayed pulse light **P2** can be made arbitrary. Therefore, the time waveform of the combined pulse light **L60**

can be adjusted. Since the attenuator does not dump the laser beam, the intensity of the laser beam can be maintained.

(37) The embodiment will be described with reference to FIG. 2 again. The combined pulse light L60 from the polarization beam splitter 61 enters the wave plate 62. The wave plate 62 is a second half-wave plate. The wave plate 62 is rotatably provided in the optical path of the combined pulse light L60. By rotating the wave plate 62 around the optical axis of the combined pulse light L60, the polarization direction of the combined pulse light L60 can be adjusted. That is, the linear polarization direction rotates according to the rotation angle of the wave plate 62.

(38) The combined pulse light L60 transmitted through the wave plate 62 is reflected by the mirror 63. The combined pulse light L60 reflected by the mirror 63 is made incident on the lens 64, the homogenizer 65, the homogenizer 66 and the lens 67 in this order.

(39) The lens 64 condenses the combined pulse light L60. The homogenizer 65 and the homogenizer 66 uniformize the spatial distribution of the combined pulse light L60. The homogenizer 65 is a combination of an array lens for dividing the beam in the X direction and an array lens for dividing the beam in the Y direction. The homogenizer 66 is a combination of an array lens for transmitting a beam divided in the X direction to a condenser lens constituting the lens 67 and an array lens for transmitting a beam divided in the Y direction to the condenser lens constituting the lens 67. The condenser lens is installed so that the beam becomes a top flat beam at an aperture stage. Thereafter, the magnification is changed by the objective lens. The lens 67 condenses the combined pulse light L60.

(40) Although not shown in Figs., the optical system 20 transmits beams in a manner such that the beam is repeatedly focused and collimated from the light source 11 to the homogenizer 65. The combined pulse light L60 may be enlarged so that the beam is divided immediately before the homogenizer 65. For example, the lens 64 may be a combination of a lens that magnifies the beam and a lens that collimates the magnified beam, specifically a telescoping lens. The lens 67 is a condenser lens for condensing a beam, and a slit (aperture) is arranged on a downstream side of the lens 67. An epi-illumination mirror for irradiating the substrate with the beam may be disposed downstream of the slit. An objective lens may be disposed downstream of the epi-illumination mirror.

(41) The combined pulse light L60 is applied to substrate 100 as the laser light L2 of FIG. 1. The laser beam L2 forms a linear irradiation region on the substrate 100. The laser beam L2 is not limited to a line beam. For example, the laser beam L2 may have a top-flat spatial distribution in both the X direction and the Y direction. On the substrate 100, the spot shape of the laser beam L2 may be a 1-mm square. Thus, the substrate 100 can be irradiated with a line beam having a uniform spatial distribution.

(42) With reference to FIG. 5, the polarization state of the laser beam L2 applied to the substrate 100 will be described. FIG. 5 is a diagram schematically showing a polarization state of a laser beam L2 including the preceding pulse beam P1 and the delay pulse beam P2. FIG. 5 shows three examples in which the angle of the wave plate 62 is changed.

(43) As shown in FIG. 5, by changing the angle of the wave plate 62, with respect to the substrate 100, the preceding pulse light P1 can be made a first linear polarization light (for example, S polarization) and the delayed pulse light P2 can be made a second linear polarization light (for example, P polarization). Alternatively, by changing the angle of the wave plate 62, with respect to the substrate 100, the preceding pulse light P1 can be made the second linear polarization light and the delay pulse light P2 can be made the first linear polarization light. That is, the polarization states of the preceding pulse light P1 and the delay pulse light P2 can be exchanged. Further, the preceding pulse light P1 and the delay pulse light P2 can be a linear polarization light inclined by, for example, 45 degrees from the first linear polarization light. Of course, the rotation angle of the wave plate 62 is not limited to 45 degree.

(44) Thus, by adjusting the rotation angle of the wave plate 62, a desired polarization state can be

achieved. Therefore, the pulse laser beam in an appropriate polarization state can be applied to the substrate **100** according to the laser irradiation process. Further, according to the laser irradiation process, a pulse laser beam having an appropriate time waveform can be applied to the substrate **100**.

(45) Note that the mirror **63** may be a partial transmission mirror which transmits a part of the combined pulse light **L60**. For example, the mirror **63** reflects 99% of the combined pulse light **L60** and transmits 1% of the combined pulse light **L60**. The combined pulse light **L60** transmitted through the mirror **63** enters the photodetector **70**. Of course, the transmittance and reflectance of the mirror **63** are not particularly limited. Further, when the detection by the photodetector **70** is not performed, the mirror **63** may have a reflectance of 100%.

(46) The photodetector **70** detects combined pulse light **L60** transmitted through the mirror **63**. The rotation angle of the half-wave plate **31** may be adjusted according to the detection result by the photodetector **70**. A photodetector **70** detects the time waveform of the combined pulse light **L60**. According to the detection result by the photodetector **70**, the rotation angle of the half-wave plate **31** may be changed so that the light amounts of the preceding pulse light **P1** and the delay pulse light **P2** become a desired ratio.

(47) The rotation angle of the wave plate **62** when the photodetector **70** detects the combined pulse light **L60** will be described. The optical properties of mirror **63** may depend on the polarization state of the incident light. For example, it is assumed that the transmittance of the mirror **63** to the S polarization is lower than the transmittance of the mirror **63** to the P polarization. In this case, the ratio of the detected light amounts of the preceding pulse light **P1** and the delayed pulse light **P2** detected by the photodetector **70** deviates from the actual light amounts of the preceding pulse light **P1** and the delayed pulse light **P2**. For example, when the preceding pulse light **P1** is S polarization and the delayed pulse light **P2** is P polarization, the photodetector **70** detects the combined pulse light **L60** with the light amount ratio in which the light amount of the P polarized delayed pulse light **P2** becomes high.

(48) Therefore, in the present embodiment, the rotation angle of the wave plate **62** is adjusted so that the S polarization component and the P polarization component in the preceding pulse light **P1** are equal to each other and the S polarization component and the P polarization component in the delay pulse light **P2** are equal to each other. Specifically, the optic axis of the wave plate **62** is arranged to incline by 22.5 degree with respect to the linear polarization light of the preceding pulse light **P1**. Thus, when the preceding pulse light **P1** passes through the wave plate **62**, the linear polarization light of the preceding pulse light **P1** is rotated by 45 degree.

(49) The delayed pulse light **P2** is linear polarization light orthogonal to the preceding pulse light **P1**. Therefore, when the delayed pulse light **P2** passes through the wave plate **62**, the linear polarization light of the delayed pulse light **P2** is also rotated by 45 degree. Therefore, the polarization ratio of the preceding pulse light **P1** incident on the mirror **63** can be made equal to the polarization ratio of the delayed pulse light **P2**. Therefore, the photodetector **70** can detect the combined pulse light **L60** more accurately. Since the time waveform of the combined pulse light **L60** can be accurately measured, the polarization state can be adjusted more appropriately. After the adjustment of the rotation angle of the half-wave plate **31** is completed, the rotation angle of the wave plate **62** may be adjusted.

(50) Further, the rotation angle of the wave plate **41** or the wave plate **53** may be adjusted based on the detection result by the photodetector **70**. The rotation angle of the wave plate **41** is changed so that the amount of light detected by the photodetector **70** becomes maximum. Thereafter, the rotation angle of the wave plate **53** is changed so that the amount of light detected by the photodetector **70** becomes maximum. Thus, the loss of the laser beam in the polarization beam splitter **61** can be reduced. After the adjustment of the rotation angles of the half-wave plate **31**, the wave plate **41**, and the wave plate **53** is completed, the wave plate **62** may be adjusted.

(51) Further, the wave plate **62** may be a quarter-wave plate. For example, in a process of

irradiating a circularly polarized laser beam with the substrate **100**, the quarter wave plate can be used as the wave plate **62**, instead of the half-wave plate.

(52) The wave plate **41** and the wave plate **53** may be omitted. The wave plate **41** or the wave plate **53** may be a quarter-wave plate, respectively.

(53) The polarization state may be controlled according to the evaluation result of the crystal state of the polysilicon film **101b**. For example, the detection unit **22** shown in FIG. **1** detects the crystal state of the polysilicon film **101b**. The rotation angles of the half-wave plate **31** and the wave plate **62** are adjusted according to the detection result of the detection unit **22**. Thus, a polarization state suitable for various process conditions can be generated. For example, the polarization state can be optimized according to the output power of the laser beam, the type of substrate **100**, the thickness of the silicon film **101**, the conveying speed of the substrate **100**, and the like. For example, the polarization state is controlled so that variation in the crystal state of the polysilicon becomes small. Therefore, the laser beam **L2** in a polarization state suitable for the annealing process can be irradiated with the substrate **100**.

(54) By using the polarization beam splitter **33** and the polarization beam splitter **61**, the loss of laser light can be reduced. The combined pulse light **L60** in a desired polarization state can be efficiently generated. That is, since a polarizer or the like is not used, absorption of the laser beam can be prevented, and the laser beam can be efficiently utilized.

(55) The laser annealing method according to the present embodiment comprises the steps of: (a) emitting a linearly polarized pulsed laser light; (b) causing the pulsed laser light to be made incident on a first half-wave plate rotatably placed; (c) branching the pulsed laser light from the first half-wave plate into a first pulsed light and a second pulsed light; (d) combining the first pulsed light with the second pulsed light, the second pulsed light being delayed from the first pulsed light by using an optical path length difference between the first pulsed light and second pulsed light; and (e) causing a combined pulsed laser to be made incident on a first wave plate rotatably provided in an optical path of the combined pulsed laser, the combined pulsed laser being generated by combining the first pulsed light with the second pulsed light at the second polarization beam splitter. By rotating the first half-wave plate and the first wave plate, a laser beam in an appropriate polarization state can be generated.

(56) The polarization state of the laser beam affects the grain size and the orientation of the crystal.

(57) In the present embodiment, the laser beam can be controlled to be in an arbitrary polarization state. Therefore, the polarization state can be controlled so as to improve the quality such as uniformity, periodicity, and orientation of the crystal. By controlling the micro crystal state, it is possible to perform an adjustment under conditions where unevenness hardly occurs.

(58) Further, as the stage **10**, it is preferable to use a flotation conveyance stage for conveying the substrate **100** while the substrate **100** is floated. In particular, the flotation conveyance stage is superior in transporting large substrates. When a large substrate is annealed, it is necessary to reduce the cost per unit product. When the large substrate is annealed using the flotation conveyance stage, the laser beam is converted into an appropriate polarization state and then irradiated with the substrate **100**. Therefore, it is possible to perform appropriate annealing. Further, by controlling the polarization state based on the evaluation result of the crystal state by the detection unit **22**, a high quality crystal can be produced with high productivity.

(59) The polarization control unit **30** can be applied to the laser irradiation apparatus other than the laser annealing apparatus. For example, the polarization control unit **30** can be incorporated in the laser irradiation apparatus of a laser machining apparatus or a laser lift-off apparatus. The laser irradiation apparatus according to the present embodiment is not limited to the above-described example.

(60) For example, in a laser machining process such as drilling, it may irradiate a circularly polarized laser beam with an object. In this case, a quarter-wave plate can be used as the wave plate **62**. The laser light source **21** generates an ultrashort pulse of laser light **L1**, and the polarization

controller **30** sets the combined pulse light **L60** to a single pulse having a desired pulse width. In the laser lift-off device, the laser beam of an ultrashort pulse may be circularly polarized and irradiated with the object.

(61) (Organic EL Display)

(62) A semiconductor device having the above-described polysilicon film is suitable for a thin film transistor (TFT) array substrate used for an organic electro-luminescence (EL) display. That is, the polysilicon film is used as a semiconductor layer having a source region, a channel region, and a drain region of a TFT.

(63) Hereinafter, a structure in which a semiconductor device according to the present embodiment is used for an organic EL display is described. FIG. **6** is a cross-sectional view of a pixel circuit of an organic EL display device illustrated in a simplified manner. The organic EL display device **300** shown in FIG. **6** is an active-matrix-type display device in which a TFT is disposed in each pixel PX.

(64) The organic EL display device **300** includes a substrate **310**, a TFT layer **311**, an organic layer **312**, a color filter layer **313**, and a sealing substrate **314**. FIG. **6** shows a top emission-type organic EL display device, in which the sealing substrate **314** side is located on the viewing side. Note that the following description is given to show an example of a configuration of an organic EL display device and this embodiment is not limited to the below-described configuration. For example, a semiconductor device according to this embodiment may be used for a bottom-emission-type organic EL display device.

(65) The substrate **310** is a glass substrate or a metal substrate. The TFT layer **311** is provided over the substrate **310**. The TFT layer **311** includes TFTs **311a** disposed in the respective pixels PX. Further, the TFT layer **311** includes wiring lines (not shown) connected to the TFTs **311a**, and the like. The TFTs **311a**, the wiring lines, and the like constitute pixel circuits.

(66) The organic layer **312** is provided over the TFT layer **311**. The organic layer **312** includes an organic EL light-emitting element **312a** disposed in each pixel PX. Further, in the organic layer **312**, separation walls **312b** for separating organic EL light-emitting elements **312a** are provided between pixels PX.

(67) The color filter layer **313** is provided over the organic layer **312**. The color filter layer **313** includes color filters **313a** for performing color displaying. That is, in each pixel PX, a resin layer colored in R (red), G (green), or B (blue) is provided as the color filter **313a**.

(68) The sealing substrate **314** is provided over the color filter layer **313**. The sealing substrate **314** is a transparent substrate such as a glass substrate and is provided to prevent deterioration of the organic EL light-emitting elements of the organic layer **312**.

(69) Electric currents flowing through the organic EL light-emitting elements **312a** of the organic layer **312** are changed by display signals supplied to the pixel circuits. Therefore, it is possible to control an amount of light emitted in each pixel PX by supplying a display signal corresponding to a display image to each pixel PX. As a result, it is possible to display a desired image.

(70) In an active matrix display device such as an organic EL display, one pixel PX is provided with one or more TFTs (for example, a switching TFT and a driving TFT). Then, the TFT of each pixel PX is provided with a semiconductor layer having a source region, a channel region, and a drain region. The polysilicon film according to the present embodiment is suitable for a semiconductor layer of a TFT. That is, by using the polysilicon film manufactured by the above manufacturing method for a semiconductor layer of a TFT array substrate, it is possible to suppress in-plane nonuniformity which is the TFT characteristics. Thus, it is possible to manufacture a display device having an excellent display characteristic with high productivity.

(71) (Manufacturing Method for Semiconductor Device)

(72) A manufacturing method for a semiconductor device using the ELA apparatus according to the present embodiment is suitable for manufacturing a TFT array substrate. A method for manufacturing a semiconductor device including a TFT will be described with reference to FIGS. **7**

and 8. FIGS. 7 and 8 are cross-sectional views showing processes for manufacturing a semiconductor device. In the following description, a manufacturing method for a semiconductor device having an inverted staggered TFT is described. FIGS. 7 and 8 shows steps for forming the polysilicon film in the manufacturing method for the semiconductor device. Note that well known methods can be used as manufacturing steps other than the polysilicon film formation, the explanations thereof are omitted.

(73) As shown in FIG. 7, a gate electrode **402** is formed over a glass substrate **401**. A gate insulating film **403** is formed over the gate electrode **402**. An amorphous silicon film **404** is formed over the gate insulating film **403**. The amorphous silicon film **404** is arranged so as to overlap the gate electrode **402** interposing the gate insulating film **403**. For example, the gate insulating film **403** and the amorphous silicon film **404** are continuously formed by a chemical vapor deposition (CVD) method.

(74) Then, the amorphous silicon film **404** is irradiated with the laser light L2 and a polysilicon film **405** is thereby formed as shown in FIG. 8. That is, the amorphous silicon film **404** is crystallized by the ELA apparatus **1** shown in FIG. 1 and the like. The polysilicon film **405** with silicon crystallized is thereby formed on the gate insulating film **403**. The polysilicon film **405** corresponds to the above polysilicon film **101b**.

(75) In addition, it has been described that the laser annealing apparatus according to the present embodiment irradiates an amorphous silicon film with laser light to form a polysilicon film in the above description, but the laser annealing apparatus may irradiate an amorphous silicon film with laser light to form a micro-crystal silicon film. Furthermore, a laser light for performing annealing is not limited to Nd:YAG laser. In addition, the method according to the present embodiment can be applied to a laser annealing apparatus that crystallizes thin films other than a silicon film. That is, as long as the laser annealing apparatus irradiates an amorphous film with laser light to form a crystallized film, the method according to the present embodiment can be applied. According to a laser annealing apparatus in accordance with the embodiment, it is possible to properly inspect a substrate with a crystallized film.

(76) From the disclosure thus described, it will be obvious that the embodiments of the disclosure may be varied in many ways. Such variations are not to be regarded as a departure from the spirit and scope of the disclosure, and all such modifications as would be obvious to one skilled in the art are intended for inclusion within the scope of the following claims.

Claims

1. A laser irradiation apparatus comprising: a laser light source configured to emit a linearly polarized pulsed laser light; a first half-wave plate rotatably provided in an optical path of the pulsed laser light; a first polarization beam splitter configured to branch the pulsed laser light from the first half-wave plate into a first pulsed light and a second pulsed light; a second polarization beam splitter configured to combined the first pulsed light with the second pulsed light, the second pulsed light being delayed from the first pulsed light by using an optical path length difference between the first pulsed light and the second pulsed light; and a first wave plate rotatably provided in an optical path of a combined pulsed light generated by combining the first pulsed light with the second pulsed light at the second polarization beam splitter, wherein the first wave plate is a second half-wave plate, and wherein the laser irradiation apparatus further comprises: a partial transmission mirror configured to extract a part of the combined pulsed light transmitted through the second half-wave plate; and a photodetector configured to detect the combined pulsed light from the partial transmission mirror.

2. A laser irradiation apparatus comprising: a laser light source configured to emit a linearly polarized pulsed laser light; a first half-wave plate rotatably provided in an optical path of the pulsed laser light; a first polarization beam splitter configured to branch the pulsed laser light from

the first half-wave plate into a first pulsed light and a second pulsed light; a second polarization beam splitter configured to combined the first pulsed light with the second pulsed light, the second pulsed light being delayed from the first pulsed light by using an optical path length difference between the first pulsed light and the second pulsed light; a first wave plate rotatably provided in an optical path of a combined pulsed light generated by combining the first pulsed light with the second pulsed light at the second polarization beam splitter; and a second wave plate rotatably provided in an optical path of a combined pulsed light from the first polarization beam splitter to the second polarization beam splitter.

3. The laser irradiation apparatus according to claim 2, wherein the second wave plate is a third half-wave plate.

4. A laser irradiation apparatus comprising: a laser light source configured to emit a linearly polarized pulsed laser light; a first half-wave plate rotatably provided in an optical path of the pulsed laser light; a first polarization beam splitter configured to branch the pulsed laser light from the first half-wave plate into a first pulsed light and a second pulsed light; a second polarization beam splitter configured to combined the first pulsed light with the second pulsed light, the second pulsed light being delayed from the first pulsed light by using an optical path length difference between the first pulsed light and the second pulsed light; and a first wave plate rotatably provided in an optical path of a combined pulsed light generated by combining the first pulsed light with the second pulsed light at the second polarization beam splitter, wherein the combined pulsed light forms a linear irradiation region along a first direction on an object.

5. The laser irradiation apparatus according to claim 4, wherein an amorphous film is formed over the object; and the film is crystalized by the combined pulsed light.

6. A laser irradiation method comprising the steps of: (a) emitting a linearly polarized pulsed laser light; (b) causing the pulsed laser light to be made incident on a first half-wave plate rotatably placed; (c) branching the pulsed laser light from the first half-wave plate into a first pulsed light and a second pulsed light; (d) combining the first pulsed light with the second pulsed light, the second pulsed light being delayed from the first pulsed light by using an optical path length difference between the first pulsed light and second pulsed light; and (e) causing a combined pulsed laser to be made incident on a first wave plate rotatably provided in an optical path of the combined pulsed laser, the combine pulsed laser being generated by combining the first pulsed light with the second pulsed light at the second polarization beam splitter, wherein the first wave plate is a second half-wave plate, and wherein the laser irradiation method further comprises the steps of: extracting a part of the combined pulsed light by causing the combined pulsed light from the second half-wave plate to be made incident on a partial transmission mirror, and detecting the combined pulsed light from the partial transmission mirror by a photodetector.

7. A laser irradiation method comprising the steps of: (a) emitting a linearly polarized pulsed laser light; (b) causing the pulsed laser light to be made incident on a first half-wave plate rotatably placed; (c) branching the pulsed laser light from the first half-wave plate into a first pulsed light and a second pulsed light; (d) combining the first pulsed light with the second pulsed light, the second pulsed light being delayed from the first pulsed light by using an optical path length difference between the first pulsed light and second pulsed light; (e) causing a combined pulsed laser to be made incident on a first wave plate rotatably provided in an optical path of the combined pulsed laser, the combine pulsed laser being generated by combining the first pulsed light with the second pulsed light at the second polarization beam splitter; and causing the second pulsed light to be made incident on a second wave plate rotatably provided in an optical path of a combined pulsed light from the first polarization beam splitter to the second polarization beam splitter.

8. The laser irradiation method according to claim 7, wherein the second wave plate is a third half-wave plate.

9. A laser irradiation method comprising the steps of: (a) emitting a linearly polarized pulsed laser light; (b) causing the pulsed laser light to be made incident on a first half-wave plate rotatably

placed; (c) branching the pulsed laser light from the first half-wave plate into a first pulsed light and a second pulsed light; (d) combining the first pulsed light with the second pulsed light, the second pulsed light being delayed from the first pulsed light by using an optical path length difference between the first pulsed light and second pulsed light; and (e) causing a combined pulsed laser to be made incident on a first wave plate rotatably provided in an optical path of the combined pulsed laser, the combined pulsed laser being generated by combining the first pulsed light with the second pulsed light at the second polarization beam splitter, wherein the combined pulsed light forms a linear irradiation region along a first direction on an object.

10. The laser irradiation method according to claim 9, wherein an amorphous film is formed over the object; and the film is crystalized by the combined pulsed light.

11. A method for manufacturing a semiconductor device, comprising the steps of: (S1) forming an amorphous film over a substrate; and (S2) annealing the amorphous film so that the amorphous film is crystalized and a crystallized film is thereby formed, wherein the step of (S2) annealing comprises the steps of: (A) emitting a linearly polarized pulsed laser light; (B) causing the pulsed laser light to be made incident on a first half-wave plate rotatably placed; (C) branching the pulsed laser light from the first half-wave plate into a first pulsed light and a second pulsed light; (D) combining the first pulsed light with the second pulsed light, the second pulsed light being delayed from the first pulsed light by using an optical path length difference between the first pulsed light and second pulsed light; and (E) causing a combined pulsed laser to be made incident on a first wave plate rotatably provided in an optical path of the combined pulsed laser, the combined pulsed laser being generated by combining the first pulsed light with the second pulsed light at the second polarization beam splitter, wherein the first wave plate is a second half-wave plate, and wherein the method further comprises the steps of: extracting a part of the combined pulsed light by causing the combined pulsed light from the second half-wave plate to be made incident on a partial transmission mirror, and detecting the combined pulsed light from the partial transmission mirror by a photodetector.

12. A method for manufacturing a semiconductor device, comprising the steps of: (S1) forming an amorphous film over a substrate; and (S2) annealing the amorphous film so that the amorphous film is crystalized and a crystallized film is thereby formed, wherein the step of (S2) annealing comprises the steps of: (A) emitting a linearly polarized pulsed laser light; (B) causing the pulsed laser light to be made incident on a first half-wave plate rotatably placed; (C) branching the pulsed laser light from the first half-wave plate into a first pulsed light and a second pulsed light; (D) combining the first pulsed light with the second pulsed light, the second pulsed light being delayed from the first pulsed light by using an optical path length difference between the first pulsed light and second pulsed light; and (E) causing a combined pulsed laser to be made incident on a first wave plate rotatably provided in an optical path of the combined pulsed laser, the combined pulsed laser being generated by combining the first pulsed light with the second pulsed light at the second polarization beam splitter, wherein the method further comprises the step of: causing the second pulsed light to be made incident on a second wave plate rotatably provided in an optical path of a combined pulsed light from the first polarization beam splitter to the second polarization beam splitter.

13. The method for manufacturing a semiconductor device according to claim 12, wherein the second wave plate is a third half-wave plate.

14. A method for manufacturing a semiconductor device, comprising the steps of: (S1) forming an amorphous film over a substrate; and (S2) annealing the amorphous film so that the amorphous film is crystalized and a crystallized film is thereby formed, wherein the step of (S2) annealing comprises the steps of: (A) emitting a linearly polarized pulsed laser light; (B) causing the pulsed laser light to be made incident on a first half-wave plate rotatably placed; (C) branching the pulsed laser light from the first half-wave plate into a first pulsed light and a second pulsed light; (D) combining the first pulsed light with the second pulsed light, the second pulsed light being delayed

from the first pulsed light by using an optical path length difference between the first pulsed light and second pulsed light; and (E) causing a combined pulsed laser to be made incident on a first wave plate rotatably provided in an optical path of the combined pulsed laser, the combine pulsed laser being generated by combining the first pulsed light with the second pulsed light at the second polarization beam splitter, wherein the combined pulsed light forms a linear irradiation region along a first direction on an object.
